THE LIVING WORLD

Biology is the science of 'life forms' and 'life processes'

Living Study

- Early men has classified all matter present on Earth in 2- categories

Animate
Living / Biotic
e.g., Plants, animals

Inanimate
Non-living / Abiotic
e.g., wind, sea, fire

- There are **two views** to understand biology :

Anthropocentric view
Here, humans are considered superior than other species
∴ only a limited progress in our biological knowledge can be seen.

Biocentric view
Here, all living species are considered equal
∴ We will try to learn about roles, features and importance of even a small species around us.

Thus, there will be immense

The World is a Family

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Even our Indian culture states that all species on Earth are like a family and are equal.

Thus, there will be immense progress in our biological knowledge.

Horizontal and Vertical similarities

- Similarities within a generation : between siblings i.e., 'Intra'generation
- Similarities between 2-generations : between parents and offsprings i.e., 'Inter'generation

- All present day living organisms are related to each other & to all organisms ever lived on Earth but to varying degrees as we all share common genetic material i.e., Nucleic acid — DNA (majority of organisms), RNA (few viruses)

Ernst Mayr

- Called as Darwin of 20th century.
- Listed among top-100 scientists of all time.
- Won TRIPLE CROWN of biology : (i) Balzan Prize (1983)
  (ii) International Prize for biology (1994)
  (iii) Crafoord Prize (1999)

- Mayr has given 'BIOLOGICAL CONCEPT OF SPECIES' which is the currently accepted definition of species.

According to this concept : The organisms in a population that can breed together and can produce fertile offsprings are said to belong to same species.

NOTE : If organisms from two different species (with

NOTE : If organisms from two different species (with similarities) try to breed together, their progeny will be sterile.

e.g.,

♂ (Male)	Cross	♀ (Female)	Offspring
Donkey (Jack)	X	Horse (Mare)	→ Mule
Horse	X	Donkey	→ Hinny
Lion	X	Tiger	→ Liger
Tiger	X	Lion	→ Tigon

All sterile offsprings

INTRODUCTION

- The living world is very diverse & it contains millions of species.
- Even some organisms are found in most extreme conditions; such organisms are called as extremophiles.
  extreme (Harsh) loving
- Ecological 'conflict' and 'cooperation' occur between organisms.
  - 'conflict' → tiger predating a deer
  - 'cooperation' → You helping your mother in households

WHAT IS LIVING ?

- Characteristic features : They include all the features found in a living organism.
  e.g., Respiration, reproduction etc.
- Defining features : These features clearly differentiates between living and non-living things.
  Such characters shall be present in all (100%) living organisms

living and non-living things.
Such characters shall be present in all (100%) living organisms without exception and shall not be present in any non-living thing (0%).
e.g., Metabolism, cellular organisation, consciousness.

GROWTH

- All organisms grow
  - Both multicellular & unicellular organisms grow by cell division.
- Twin Characters
  - ↑ in mass
  - ↑ in no. of individuals

Unicellular

In-vitro culture i.e., Outside the body of organism

single bacteria (invisible) → (8-10 hours) → Bacterial colony ($10^6 - 10^7$ bacteria)

Clearly visible

- Growth, cell division and reproduction are mutually inclusive events in unicellular organisms as all these events occur simultaneously.
- In multicellular organisms, growth and reproduction are mutually exclusive events.

Growth

- Plants grow continuously throughout their life
- Animals grow upto a certain age; division occur in certain tissues to replace

throughout their life

age; division occur in certain tissues to replace lost cells.

∵ Increase in mass is also considered as growth

∴ Non-living things such as mountains, sand mounds and boulders can grow by accumulation of matter (accretion).

∴ Growth is not a defining feature of living organisms.

- In living organisms, growth is from inside (intususception)
  ↓ Defining
- Dead organisms do not grow.

REPRODUCTION

Again — To produce

- It is biological process by which new organisms (progenies) are produced by parents more or less similar to them.
  - more → Organisms produced, are of same species
  - less similar → Due to variations among parents & offsprings.

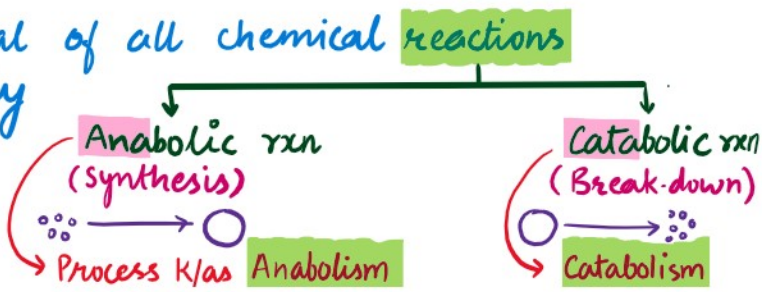
Type of asexual reproduction	Possessed by organisms
Millions of asexual spores produced	Fungi
Budding	Yeast, Hydra
True regeneration	Planaria (flat worm)
Fragmentation	Fungi, Filamentous algae, Protonema of mosses

∵ Some organisms, like mules, infertile human couples (due to abnormalities) and worker bees (∵ among ♀-bees, only queen can reproduce) do not reproduce.

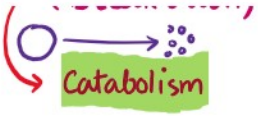
∴ Reproduction is also not a defining feature.

METABOLISM

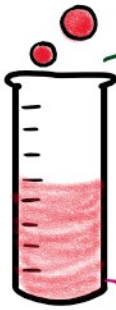
Metabolism refers to sum-total of all chemical reactions







NOTE :→ Catabolic reactions involves bond breaking (thus they release energy and hence they are exothermic or exergonic).
→ Anabolic reactions need energy, hence they are endothermic or endergonic.



→ Metabolic reactions can be Performed "in-vitro" ⟶ Such reactions are similar to reactions in our body
∴ They are living reactions.

→ However, such 'in-vitro' reactions are neither living nor non-living when tried to classify.

- Metabolism and cellular organisation are defining features of living.

Consciousness and Self-consciousness

Technically complicated and most obvious feature; It is the ability of an organism to be 'aware of its surrounding'.

Only shown by humans, as only humans are aware of themselves.
↓
Patients lying in coma with no self consciousness are living and are called BRAIN-DEAD.

SENSE

Sense environmental changes
(like we do by our

RESPONCE

Respond to environmental stimulus

changes (like we do by our sense organs)

'stimulus
- Physical (light)
- Chemical (Acids, Bases)
- Biological (Organisms)

- Photoperiod (light, duration) affects reproduction in seasonal breeders, both Plants and animals
  - Plants shows flowering based on duration of sunlight
  - Animals get to know about changes in seasons by sensing the relative duration of sunlight.

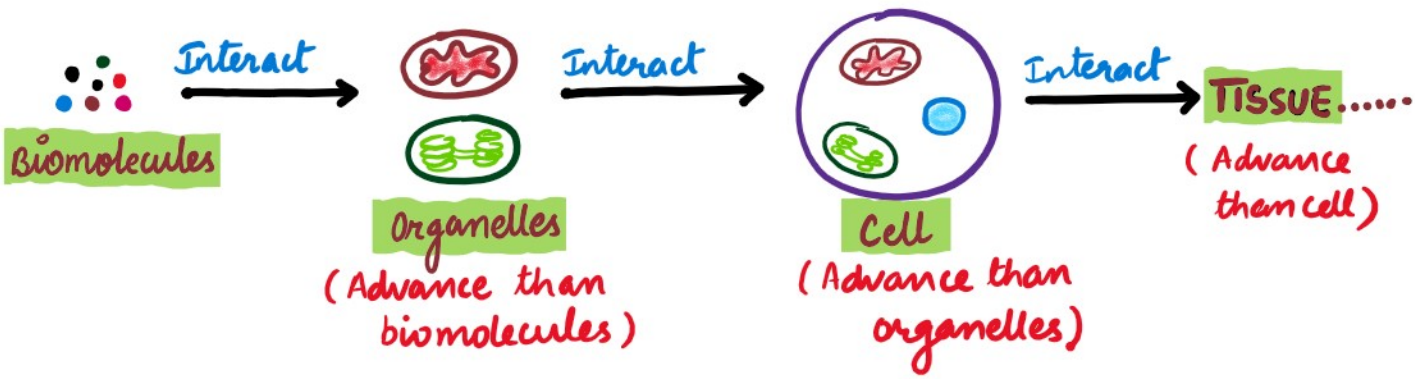
∴ Consciousness is a defining feature of living organisms.

NOTE: Self-consciousness is not a defining feature.

Interaction and Emergence

All living phenomena are due to underlying interactions.

Interactions at lower level always give rise to emerging / advance properties at higher level.



Definition of living Organisms

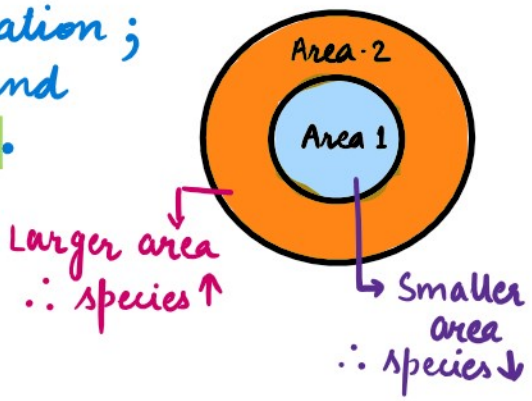
Definition of Living Organisms

- They are self-replicating, evolving, self-regulating, interactive systems capable of responding to external stimuli.

WHAT IS LIVING?

- The living world includes millions of species of organisms like that of plants, animals and microbes.

- When we increase our area of observation; the no. of species that we will found in a larger area will also increase.



- New species are obviously found when we explore new areas but are found even when we explore old areas.

- Every different kind of organism that we see, represents a species and we have known and described 1.7 - 1.8 million (17-18 lakh) species till date.

This is our 'Biodiversity'

- Each of these species is known by its local / vernacular names which vary from place to place.
∴ Creates confusion e.g., Calotropis is called : Aak (in Rajasthan)
Aakda (in Haryana)
Madaar (in M.P)

- To avoid such confusions, standardization of names is important so that an organism shall be known by same name all over the world.

- The process of naming different organisms is 'Nomenclature'

world.

- The process of naming different organisms is 'Nomenclature'. But before naming an organism, the organism need to be described correctly; this process is called 'Identification'.

Organisms	Authority and Rules for their Naming
Plants	ICBN: International Code for Botanical Nomenclature
Animals	ICZN: International Code of Zoological Nomenclature
Bacteria	ICNB: International Code of Nomenclature of Bacteria
Virus	ICVN: International Code of virus Nomenclature
Cultivated Plants	ICNCP: International Code of Nomenclature for cultivated Plants

BINOMIAL NOMENCLATURE

Bi : (2) → 1st component is Genus / Generic name
→ 2nd component is species / Specific epithet

Nomial : components of names

Nomenclature : Given by Linnaeus

Rules : Name of mango is written as : Mangifera indica Linn.

- They are Latinised or derived from Latin irrespective of their origin

Mangifera (1st alphabet of Genus is capital) **indica** (1st alphabet of species is lowercase) Linn. / *Mangifera indica* Linn. (Typed in italics to show their 'Latin origin')

genus is capital

lowercase

Seperately underlined
when hand-written

their 'Latin origin'

Name of author in abbrevia-ted form at the end of the sps. name (in Roman script)

∴ Not italicised or underlined

There are millions of species on Earth, thus to understand and study them in a better way - we need to classify them into convinient categories based on easily observable characters.

- These categories are also called: **Group / Rank / Unit / Taxa** (can represent different levels)

e.g., Humans belongs to order - primata;
class - mammalia; Kingdom - Animalia

∴ These **class**, **order** and **kingdom** are all taxa but at **different levels**.

∴ All organisms need to be classified in various 'Taxa'. This process is called **Taxonomy**.

Modern Taxonomy

Based on : (i) Developmental process
(ii) Ecological information
(iii) Cell structure
(iv) Internal str. (Anatomy)
(v) External str. (Morphology)

Conventional Taxonomy

Based on : (i) Characterisation
(ii) Identification
(iii) Classification
(iv) Nomenclature

SYSTEMATICS = Taxonomy + Evolutionary / Phylogenetic relations

'Systema' - It is a 'Latin' word which means 'systematic arrangement'
Linnaeus used 'Systema Naturae' title for his book.

Let's say I went to a jungle and found a new species, what will I do as a biologist.

1. **Characterisation** : Initially, I would note all the characters of this organism like here : chitinous exoskeleton, joint appendages, presence of antenna.

2. **Identification** : [illegible]



Periplanata americana

2. **Identification** : Now, since I know some characters of these organisms, I will get an idea of what it is. e.g., in this case, it seems to be an insect.

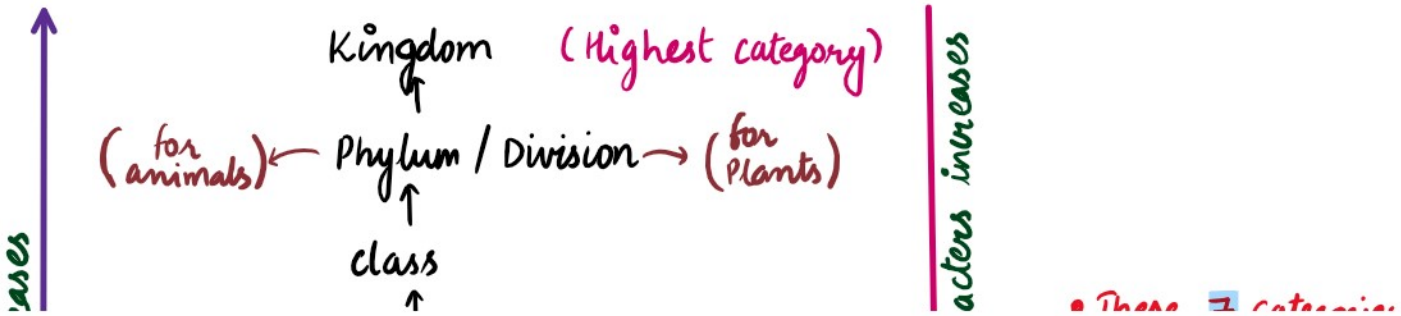
3. **Classification** : Based on its characters, the organism is classified to whatever level possible. In this case, the new organism is similar to cockroach upto genus level.
∴ Kingdom, phylum, class, order, family and genus of new organism is same as cockroach.

∵ We need to allot Kingdom, Phylum, Class..... species to a newly identified organism. ∴ We can say that classification is a multistep process where each step represents a - group / category / rank / Taxa / unit of classification.

4. **Nomenclature** : ∵ genus is already named, only the species has to be named.

TAXONOMIC CATEGORIES

- When we arrange all taxonomic categories in ascending or descending sequence - it is called as - 'Taxonomic Heirarchy'.



- These 7 categories

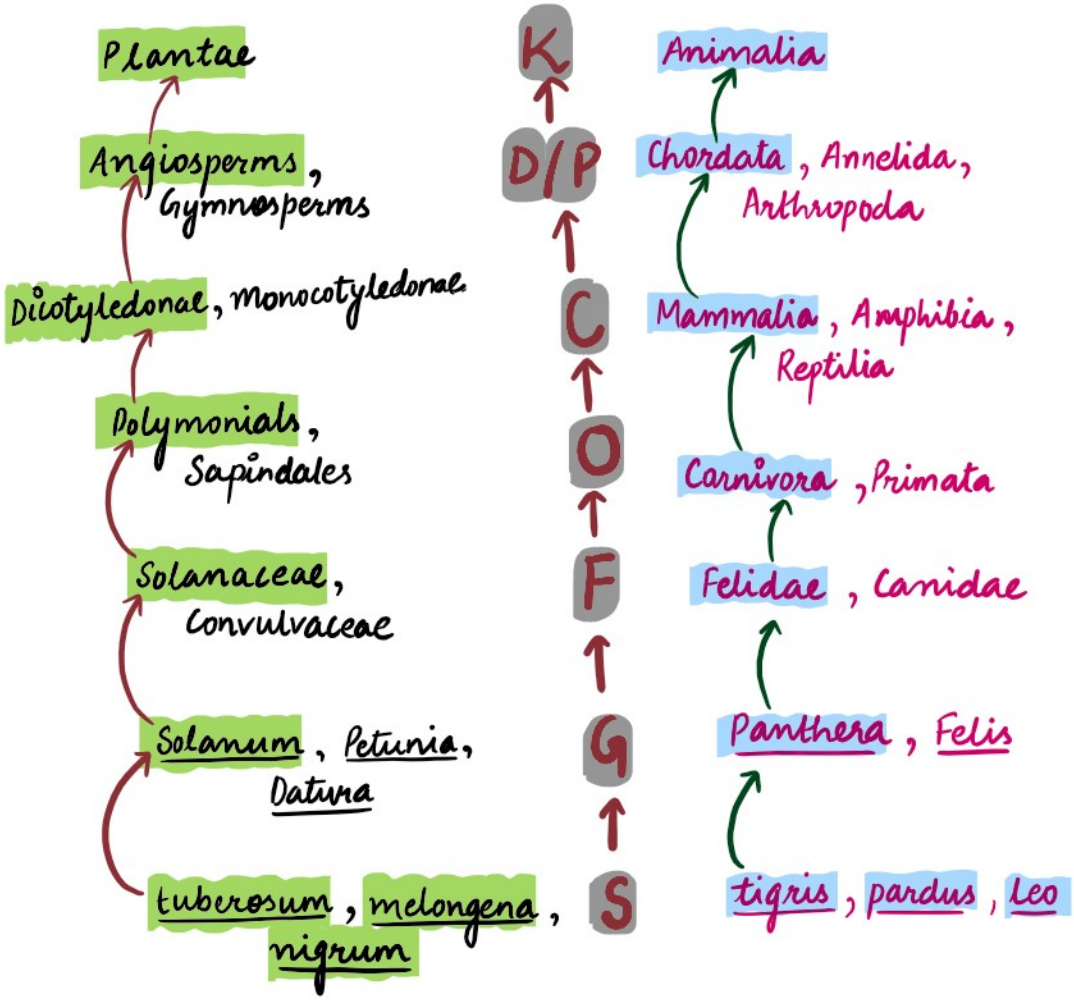
No. of organism increases

class
↑
order
↑
Family
↑
Genus
↑
Species (lowest category)

No. of common characters

• These 7 categories are called 'obligate categories'.

e.g.,



• These taxa are distinct biological entities (i.e., contain many living organisms) not only morphological aggregates (means the organisms are kept in a taxa not only on the basis of morphological characters but a lot of other features are taken into account).

Species : Group of organisms with fundamental similarities.

Genus : Group of related species ; Species present in a genus are more similar to each other as compared to species of some other genus.

- **Monotypic**
  - contain 1 species
  - e.g., Homo sapiens
- **Polytypic**
  - contain >1 species
  - e.g., Panthera leo, P. tigris, P. pardus

Family: a group of related genera with still less number of similarities as compared to genus and species.
Among plants, families are characterised on the basis of 'BOTH' vegetative and reproductive features.

e.g., Solanum, Petunia, Datura → Solanaceae family
Panthera, Felis → Felidae family

Order : Generally, order and other higher categories are identified based on the aggregates (average) of characters.
e.g., Plant families like Convolvulaceae, Solanaceae are included in the order Polymoniales mainly based on the floral characters.

- Animal order Carnivora includes families like Felidae & Canidae.

Class : includes related orders.
e.g., Primata (monkey, gorilla, gibbon) → Class Mammalia

Class :

e.g., Primata (monkey, gorilla, gibbon) → Class Mammalia
↓ Order
↑ Carnivora (tiger, dog, cat)

Phylum : Based on the common features like presence of notochord & dorsal hollow neural system, are included in phylum Chordata (Class - Mammalia, Reptilia, Amphibia, Aves, Pisces).

NOTE : In case of plants, classes with a few similar characters are assigned to a higher category called Division.

Common Name	Species	Genus	Family	Order	Class	Phylum/ Division
Man	sapiens	Homo	Hominidae	Primata	Mammalia	Chordata
Housefly	domestica	Musca	Muscidae	Diptera	Insecta	Arthropoda
Mango	indica	Mangifera	Anacardiaceae	Sapindales	Dicotyledon-ae	Angiospermae
Wheat	aestivum	Triticum	Poaceae	Poales	Monocotyle-donae	Angiospermae